Lab 2 – Responsive Navbar and Grid Drill

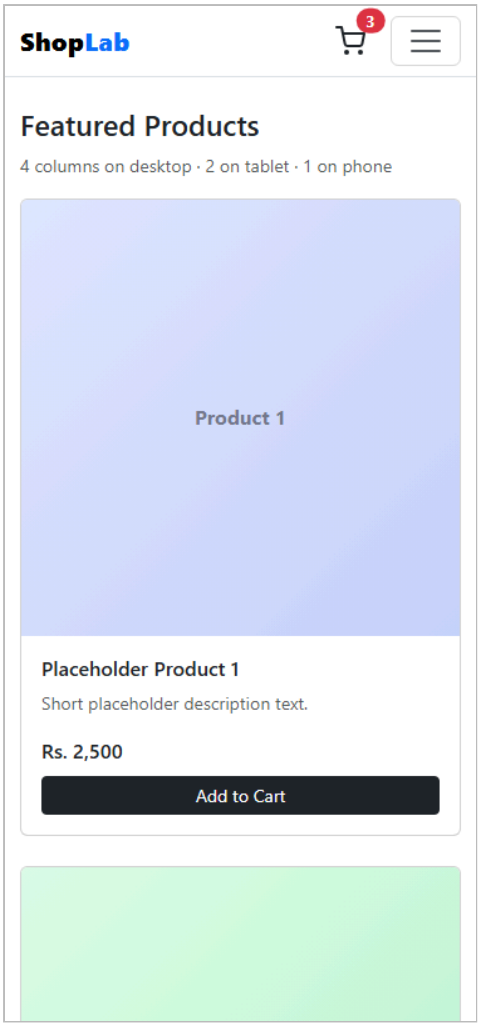
Name: Muhammad Hammad Sarwar **Registration No.:** FA24-BCS-068

Folder: `lab-2/` · Files: `index.html`, `style.css`, Bootstrap 5.3.3 from the jsDelivr CDN, `screenshots/`, this PDF.

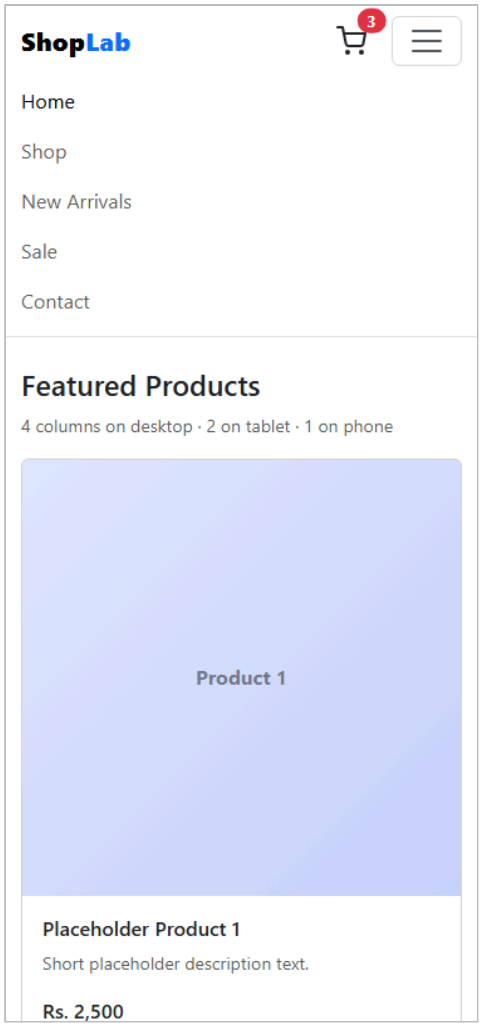
Requirements checklist

Requirement	How it is done	Status
Responsive navbar: logo left, links center, cart icon right	Bootstrap `navbar`. Logo "ShopLab" on the left, 5 links in the exact centre (in `style.css` the logo and the cart area each get `flex: 1 1 0`), cart icon with a badge on the right (`order-md-last`).	**Done**
Collapses into a hamburger below 768px	`navbar-expand-md`: Bootstrap's md breakpoint is 768px, so from 767px down the links hide inside `.collapse` and the hamburger button (`data-bs-toggle="collapse"`) opens them. The logo and cart stay visible. Checked in the browser: hamburger at 767px, links shown at 768px.	**Done**
Product grid: 4 columns desktop / 2 tablet / 1 phone	`row row-cols-1 row-cols-sm-2 row-cols-lg-4 g-4` with 8 placeholder product cards: 1 column below 576px (phone), 2 columns from 576px (tablet), 4 columns from 992px (desktop). Checked: 1 per row at 375px, 2 at 768px, 4 at 1280px.	**Done**
Flexbox or Grid plus Bootstrap or Tailwind	Bootstrap 5.3.3 (CDN with integrity hashes). Bootstrap's navbar and grid rows are Flexbox; the cards use `d-flex flex-column` so the buttons line up at the bottom.	**Done**
Placeholder content	Gradient placeholder boxes instead of photos, placeholder names, text and prices.	**Done**
Screenshots at phone / tablet / desktop	375px (menu closed and open), 768px and 1280px – next pages.	**Done**
Extra	W3C validator: 0 errors, 0 warnings. No horizontal scrolling at any width. Only Bootstrap's own JavaScript is used (for the hamburger).	**Done**

Screenshots – phone (375px)

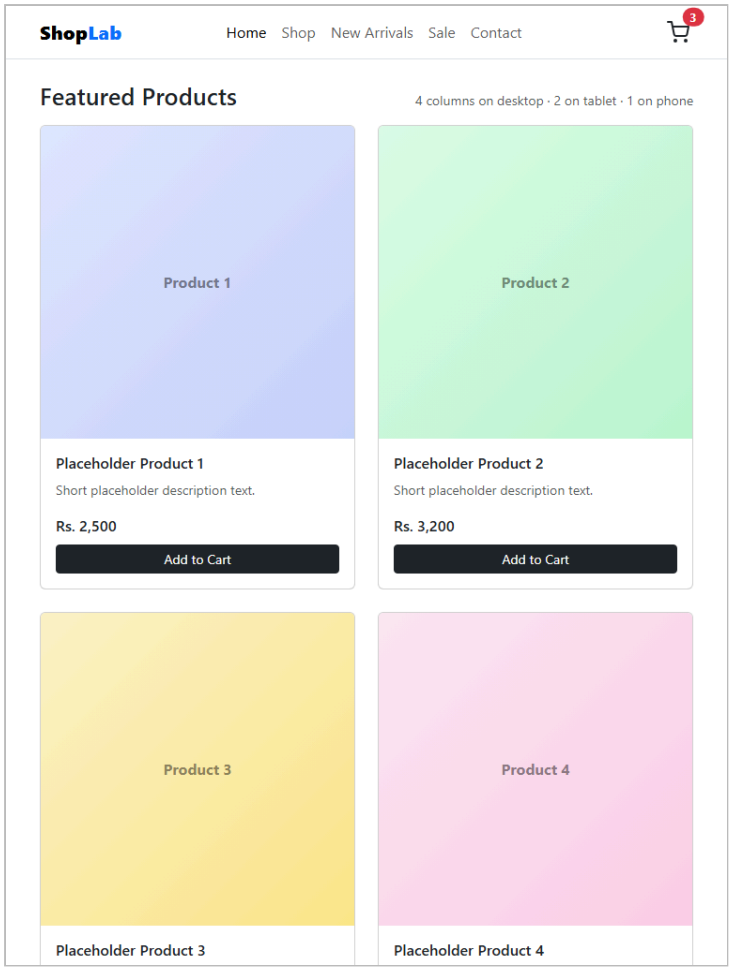


375px – hamburger, 1 column

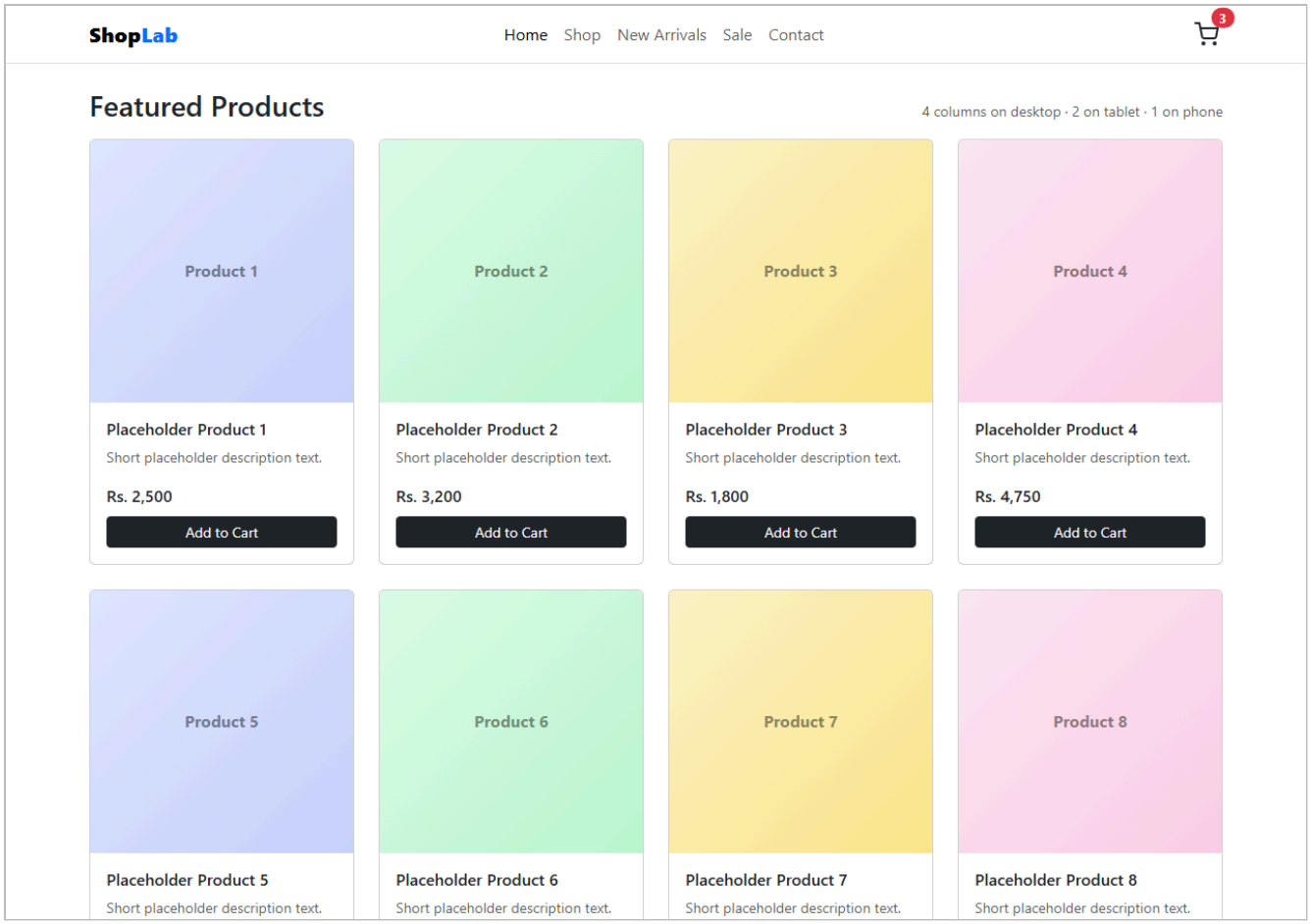


375px – hamburger menu open

Screenshots – tablet (768px) and desktop (1280px)



768px (tablet) – links shown, 2 columns



1280px (desktop) – logo left, links centre, cart right, 4 columns

Code – index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <meta name="description" content="Lab 2 - responsive navbar and product grid drill with Bootstrap 5.">
    <title>Lab 2 – Responsive Navbar and Grid Drill</title>

    <!-- Bootstrap 5.3.3 (framework) + my own small stylesheet -->
    <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
        integrity="sha384-QWTKZyjpPEjISv5WaRU9OFeRpok6YctnYmDr5pNlyT2bRjXh0JMhjY6hW+ALEwIH" crossorigin="anonymous">
    <link rel="stylesheet" href="style.css">
</head>

<!--
    Lab 2 - Responsive Navbar and Grid Drill
    Muhammad Hammad Sarwar - FA24-BCS-068

    Navbar: logo left, links centre, cart icon right.
            navbar-expand-md -> below 768px the links collapse into a hamburger menu.
    Grid:   row-cols-1 row-cols-sm-2 row-cols-lg-4
            -> 1 column on phones, 2 on tablets, 4 on desktops (Bootstrap grid = Flexbox).
-->

<body>

    <!-- ============ RESPONSIVE NAVBAR ============ -->
    <header>
        <nav class="navbar navbar-expand-md bg-white border-bottom sticky-top" aria-label="Main">
            <div class="container">

                <!-- 1. Logo (left) -->
                <a class="navbar-brand brand-logo" href="index.html">Shop<span>Lab</span></a>

                <!-- 3. Cart icon + hamburger (right). order-md-last puts this after the links on desktop -->
                <div class="nav-actions d-flex align-items-center order-md-last">
                    <a class="cart-link position-relative" href="#cart" aria-label="Cart, 3 items">
                        <svg viewBox="0 0 24 24" width="24" height="24" fill="none" stroke="currentColor"
                            stroke-width="2" stroke-linecap="round" stroke-linejoin="round" aria-hidden="true">
                            <circle cx="9" cy="21" r="1"></circle>
                            <circle cx="20" cy="21" r="1"></circle>
                            <path d="M1 1h4l2.68 13.39a2 2 0 0 0 2 1.61h9.72a2 2 0 0 0 2-1.61L23 6H6"></path>
                        </svg>
                        <span class="position-absolute top-0 start-100 translate-middle badge rounded-pill bg-
danger">3</span>
                    </a>

                    <!-- Hamburger: only shown below 768px (Bootstrap's own collapse script opens the menu) -->
                    <button class="navbar-toggler ms-3" type="button" data-bs-toggle="collapse"
                        data-bs-target="#mainNav" aria-controls="mainNav" aria-expanded="false"
                        aria-label="Toggle navigation">
                        <span class="navbar-toggler-icon"></span>
                    </button>
                </div>

                <!-- 2. Links (centre on desktop, collapsible list on phones) -->
                <div class="collapse navbar-collapse" id="mainNav">
                    <ul class="navbar-nav mx-auto">
                        <li class="nav-item"><a class="nav-link active" aria-current="page" href="#">Home</a></li>
                        <li class="nav-item"><a class="nav-link" href="#products">Shop</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">New Arrivals</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">Sale</a></li>
                        <li class="nav-item"><a class="nav-link" href="#">Contact</a></li>
                    </ul>
                </div>
            </div>
        </nav>
    </header>

    <main class="container py-4">

        <!-- ============ RESPONSIVE PRODUCT GRID ============ -->
        <section id="products" aria-labelledby="products-heading">
            <div class="d-flex flex-wrap justify-content-between align-items-baseline gap-2 mb-3">
```

```
    <h1 id="products-heading" class="h3 mb-0">Featured Products</h1>
    <p class="text-body-secondary small mb-0">4 columns on desktop · 2 on tablet · 1 on phone</p>
</div>

<!-- 1 column (phone) / 2 columns from 576px (tablet) / 4 columns from 992px (desktop) -->
<div class="row row-cols-1 row-cols-sm-2 row-cols-lg-4 g-4">
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-1" role="img" aria-label="Placeholder image">Product 1</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 1</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 2,500</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-2" role="img" aria-label="Placeholder image">Product 2</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 2</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 3,200</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-3" role="img" aria-label="Placeholder image">Product 3</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 3</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 1,800</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-4" role="img" aria-label="Placeholder image">Product 4</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 4</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 4,750</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-1" role="img" aria-label="Placeholder image">Product 5</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 5</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 2,950</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-2" role="img" aria-label="Placeholder image">Product 6</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 6</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                <p class="fw-semibold mt-auto mb-2">Rs. 5,400</p>
                <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
            </div>
        </article>
    </div>
    <div class="col">
        <article class="card h-100 product-card">
            <div class="product-img ph-3" role="img" aria-label="Placeholder image">Product 7</div>
            <div class="card-body d-flex flex-column">
                <h2 class="h6 card-title">Placeholder Product 7</h2>
                <p class="card-text text-body-secondary small">Short placeholder description text.</p>
```

```
                            <p class="fw-semibold mt-auto mb-2">Rs. 1,250</p>
                            <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
                        </div>
                    </article>
                </div>
                <div class="col">
                    <article class="card h-100 product-card">
                        <div class="product-img ph-4" role="img" aria-label="Placeholder image">Product 8</div>
                        <div class="card-body d-flex flex-column">
                            <h2 class="h6 card-title">Placeholder Product 8</h2>
                            <p class="card-text text-body-secondary small">Short placeholder description text.</p>
                            <p class="fw-semibold mt-auto mb-2">Rs. 3,600</p>
                            <a href="#" class="btn btn-dark btn-sm w-100">Add to Cart</a>
                        </div>
                    </article>
                </div>
            </div>
        </section>
    </main>

    <footer class="border-top py-3 text-center small text-body-secondary">
        Lab 2 – Muhammad Hammad Sarwar (FA24-BCS-068)
    </footer>

    <!-- Bootstrap bundle: only needed for the hamburger collapse -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"
        integrity="sha384-YvpcrYf0tY3lHB60NNkmXc5s9fDVZLESaAA55NDzOxhy9GkcIdslK1eN7N6jIeHz"
        crossorigin="anonymous"></script>
</body>

</html>
```

Code – style.css

```
/* ========================================================================
   Lab 2 - my own styles on top of Bootstrap 5.3.3
   Bootstrap does the responsive work (navbar-expand-md, row-cols-*);
   this file only adds the logo, the centred links and the placeholder images.
   ======================================================================== */

/* Logo */
.brand-logo {
    font-weight: 800;
    letter-spacing: 0.5px;
}

.brand-logo span {
    color: #0d6efd;
}

/* Cart icon */
.cart-link {
    color: #212529;
    display: inline-flex;
    padding: 4px;
}

.cart-link:hover {
    color: #0d6efd;
}

/* From 768px (when the links are shown): logo and cart area take equal space
   on both sides, so the links sit exactly in the centre of the navbar */
@media (min-width: 768px) {
    .navbar .brand-logo,
    .navbar .nav-actions {
        flex: 1 1 0;
    }

    .navbar .nav-actions {
        justify-content: flex-end;
    }

    .navbar .navbar-collapse {
        flex-grow: 0;
    }
}

/* Below 768px the opened menu gets a little space above it */
@media (max-width: 767.98px) {
    .navbar-collapse .navbar-nav {
        padding-top: 0.5rem;
    }
}

/* Placeholder product images (no real photos needed for this drill) */
.product-img {
    aspect-ratio: 1 / 1;
    display: flex;
    align-items: center;
    justify-content: center;
    font-weight: 700;
    color: rgba(0, 0, 0, 0.45);
    border-top-left-radius: var(--bs-card-inner-border-radius);
    border-top-right-radius: var(--bs-card-inner-border-radius);
}

.ph-1 { background: linear-gradient(135deg, #e0e7ff, #c7d2fe); }
.ph-2 { background: linear-gradient(135deg, #dcfce7, #bbf7d0); }
.ph-3 { background: linear-gradient(135deg, #fef3c7, #fde68a); }
.ph-4 { background: linear-gradient(135deg, #fce7f3, #fbcfe8); }

.product-card {
    transition: transform 0.15s ease, box-shadow 0.15s ease;
}

.product-card:hover {
    transform: translateY(-3px);
```

```
    box-shadow: 0 6px 18px rgba(0, 0, 0, 0.08);
}
```